

Exercise 3

- ★ Consider electron capture

$$A(J = 0) + e^- \rightarrow B^*(J = 1) + \nu_e,$$

where the captured electron is in an s -state. Choose the z -axis along the neutrino momentum.

- List the allowed spin projections of the neutrino and of B^* .
- Show that the spin projection $S_z(B^*)$ tags the neutrino helicity.
- Explain why this statement becomes ambiguous if the quantization axis is chosen independently of the neutrino momentum.

Exercise 4

- ★ Suppose the excited nucleus B^* recoils with speed $v \ll c$ and emits a photon of energy E_γ in its rest frame.

- Show that in the laboratory frame, to first order in v/c

$$E_\gamma^{\text{lab}} \simeq E_\gamma \left(1 + \frac{v}{c} \cos \theta \right),$$

where θ is the emission angle with respect to the recoil direction.

- Which emission direction maximizes the photon energy?
- Explain why resonant absorption preferentially selects photons emitted approximately along the recoil direction of B^* .